MSML610: Advanced Machine Learning

Lesson 09.3: Multi-armed Bandits

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References:

- ***Introduction***

What are Multi-Armed Bandits?

- Imagine you're in a casino facing K slot machines (aka "one-armed bandits")
  - Each machine has a different unknown probability of winning
  - Limited budget of plays
  - *Goal*: Maximize total winnings
- **Multi-Armed Bandit (MAB)** is a sequential decision-making problem where:
  - At each time step $t = 1, 2, \ldots, T$, choose one action (arm) from K options
  - After choosing arm i, receive a random reward $r_t \sim P_i$
  - Reward distributions $P_1, \ldots, P_K$ are unknown
  - Learn which arms are best while collecting rewards
- **Key Challenge**: Balance:
  - *Exploration*: Try different arms to learn their reward distributions
  - *Exploitation*: Play the arm you believe is best

Why Study Bandits?

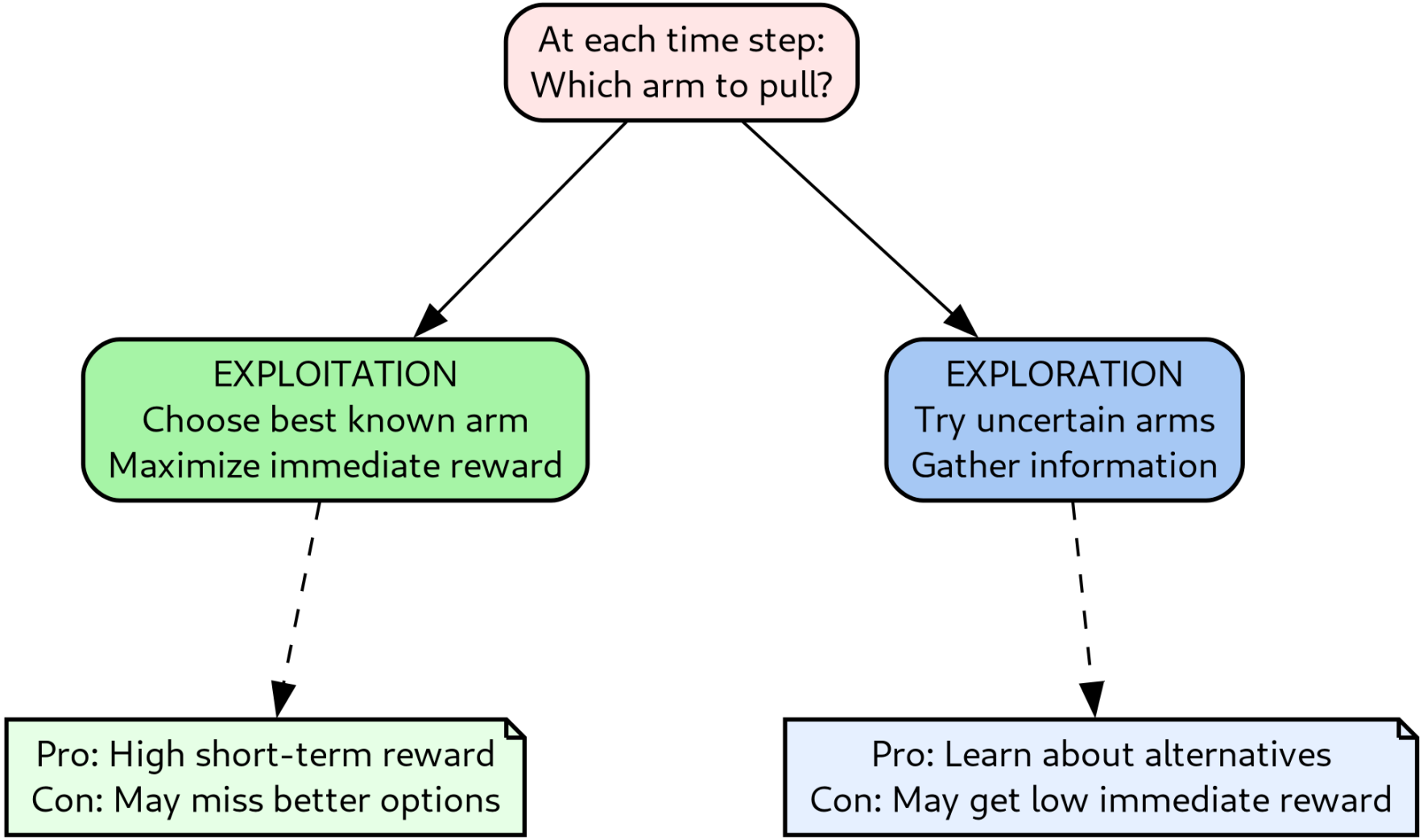
- **Fundamental Problem in Sequential Decision-Making**
  - Core model for learning under uncertainty
  - Simplest non-trivial reinforcement learning setting
  - Rich theoretical framework with practical algorithms
- **Tractable Analysis**
  - Stateless: No long-term consequences of actions
  - Clean mathematical formulation
  - Provable regret bounds and optimality results
  - Serves as building block for more complex RL problems
- **Wide Applicability**
  - Online advertising (which ad to show?)
  - Recommendation systems (which movie to suggest?)
  - Clinical trials (which treatment to test?)
  - Website optimization (A/B testing)
  - Resource allocation (which server to query?)

Applications: Real-World Decision-Making

- Sequential decisions with uncertain outcomes and limited feedback
- **Online Advertising**
  - Platform must choose which ad to display
  - Each ad click generates revenue
  - Must learn which ads perform best
  - Trade-off: Show proven ads vs. test new ones
- **Clinical Trials**
  - Allocate patients to treatment arms
  - Each treatment has unknown efficacy
  - *Ethical imperative*: Minimize patients receiving inferior treatments
  - *Exploration-exploitation* is life-or-death
- **A/B Testing**
- **Recommendation Systems**
  - Netflix/Spotify: Which content to recommend?
  - Each item has unknown appeal to user
  - Must balance popular items vs. discovery
  - Learn user preferences over time
- **Network Routing**
  - Choose which server/path to route request
  - Each option has variable latency/reliability
  - Conditions change over time
  - Need fast, adaptive decisions
- **Financial Portfolio Allocation**
  - Allocate capital among investment options
  - Each asset has unknown

The Exploration–Exploitation Tradeoff

- The "Exploration–Exploitation Tradeoff" is the central dilemma of sequential learning



- **Why This Tradeoff is Unavoidable:**

The K-Armed Bandit Problem: Formal Definition

- **Setting**:
  - K arms (actions), indexed $i = 1, \ldots, K$
  - Time horizon: T rounds (sometimes $T = \infty$)
  - At round $t = 1, \ldots, T$:
    1. Agent selects arm $A_t \in \{1, \ldots, K\}$
    2. Environment reveals reward $R_t \sim P_{A_t}$
    3. Agent observes only R_t (not rewards from other arms)
- **Stochastic Bandit Assumption**:
  - Each arm i has a fixed reward distribution P_i with mean $\mu_i \triangleq \mathbb{E}[R|A = i]$
  - Rewards are independent across rounds: $R_t \perp R_{t'} | A_t, A_{t'}$ for $t \neq t'$
  - The means $\mu_1, \ldots, \mu_K$ are unknown to the agent
- **Goal**: Design a policy (algorithm) that selects $A_1, A_2, \ldots, A_T$ to maximize

$$\text{Total Reward} = \sum_{t=1}^{T} R_t$$

- **Key Insight**: Can't maximize reward with certainty (rewards are random), but can minimize **regret**

Reward Distributions

- Random rewards
- Expected reward
- Unknown parameters

Performance Metrics

- Optimal arm
- Regret
- Cumulative regret

Regret Lower Bounds

- Why lower bounds matter
- Lai–Robbins bound
- Implications for algorithm design

Baseline Strategies

- Greedy algorithm
- Random selection
- Limitations of naive approaches

ϵ-Greedy Algorithm

- Algorithm description
- Exploration schedules
- Regret behavior

Optimism in the Face of Uncertainty

- Core idea
- Why optimism works
- Confidence intervals

Upper Confidence Bound (UCB)

- UCB1 algorithm
- Arm selection rule
- Key intuition

Regret Analysis of UCB

- Logarithmic regret
- Role of concentration inequalities
- Comparison to ϵ-greedy

Variants of UCB

- UCB-V
- KL-UCB
- When variants matter

Bayesian Bandits

- Bayesian modeling of rewards
- Priors and posteriors
- Bayesian decision-making

Thompson Sampling

- Posterior sampling idea
- Algorithm steps
- Practical performance

Comparison of UCB and Thompson Sampling

- Frequentist vs. Bayesian views
- Empirical vs. theoretical tradeoffs
- When to use each

Adversarial Bandits

- Motivation for adversarial models
- Difference from stochastic bandits
- Adversary assumptions

EXP3 Algorithm

- Probability-weighted arm selection
- Importance weighting
- Regret guarantees

Stochastic vs. Adversarial Settings

- Model assumptions
- Algorithm robustness
- Use cases

Contextual Bandits

- Using side information
- Feature vectors
- Decision rules with context

Linear Contextual Bandits

- Linear reward model
- Parameter estimation
- Assumptions

LinUCB

- Algorithm structure
- Confidence ellipsoids
- Regret guarantees

Contextual Thompson Sampling

- Bayesian linear models
- Posterior sampling with context
- Comparison to LinUCB

Bandits vs. Reinforcement Learning

- Stateless vs. stateful problems
- Bandits as a special case of RL
- When RL is needed

Non-Stationary Bandits

- Changing reward distributions
- Concept drift
- Practical motivation

Algorithms for Non-Stationary Bandits

- Discounted UCB
- Sliding-window methods
- Restart strategies

Structured Bandits

- Linear bandits
- Combinatorial bandits
- Graph and rank-one structures

Best-Arm Identification

- Pure exploration setting
- Fixed-confidence vs. fixed-budget
- Sample complexity

Practical Issues

- Delayed feedback
- Offline evaluation
- Safety constraints